QUICKCHECK
FROM EXAM TO LEARNING INSIGHT

SAMPLE NATIONAL HIGH SCHOOL

# Item Analysis

Science 7 – Quarter 1 Quiz

| CLASS | LEARNING AREA | TERM · SCORE COLUMN |
|---|---|---|
| Grade 7 – Sampaguita (Simple scorebook) | Science | TERM1 · WW1 |
| **NUMBER OF ITEMS** | **LEARNERS TESTED** | **SCHOOL YEAR** |
| 20 | 1 | 2026–2027 |
| **PAPERS CHECKED** | **DATE OF REPORT** | |
| November 06, 2026 | November 06, 2026 | |

## I. Item Analysis

**Table 1.** Summary of Item Statistics

| Measure | Value |
|---|---|
| Reliability (KR-20) | Not computable |
| Upper and lower groups (27% each) | 1 learner each |
| Items rated Very Easy | 19 |
| Items rated Easy | 0 |
| Items rated Average | 0 |
| Items rated Difficult | 0 |
| Items rated Very Difficult | 1 |
| Items to retain | 0 |
| Items to revise | 0 |
| Items to reject | 20 |

*Note.* Fewer than 20 learners took the test, so the discrimination indices rest on small groups and should be read with caution.

**Table 2.** Difficulty Index, Discrimination Index, and Decision per Item

| Item | Key | Correct (f) | p | Difficulty | D | Discrimination | Decision | Most Chosen Wrong Option |
|---|---|---|---|---|---|---|---|---|
| 1 | B | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |
| 2 | D | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |
| 3 | A | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |
| 4 | C | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |
| 5 | C | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |
| 6 | A | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |

| Item | Key | Correct (f) | p | Difficulty | D | Discrimination | Decision | Most Chosen Wrong Option |
|---|---|---|---|---|---|---|---|---|
| 7 | B | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |
| 8 | D | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |
| 9 | B | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |
| 10 | A | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |
| 11 | D | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |
| 12 | C | 0 | 0.00 | Very Difficult | 0.00 | Poor | **Reject** | D (1) |
| 13 | C | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |
| 14 | B | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |
| 15 | A | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |
| 16 | D | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |
| 17 | B | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |
| 18 | C | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |
| 19 | A | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |
| 20 | D | 1 | 1.00 | Very Easy | 0.00 | Poor | **Reject** | — |

*Note.* p = difficulty index; D = discrimination index. Correct (f) = number of learners who answered correctly.

**Figure 1.** Percentage of Learners Answering Each Item Correctly

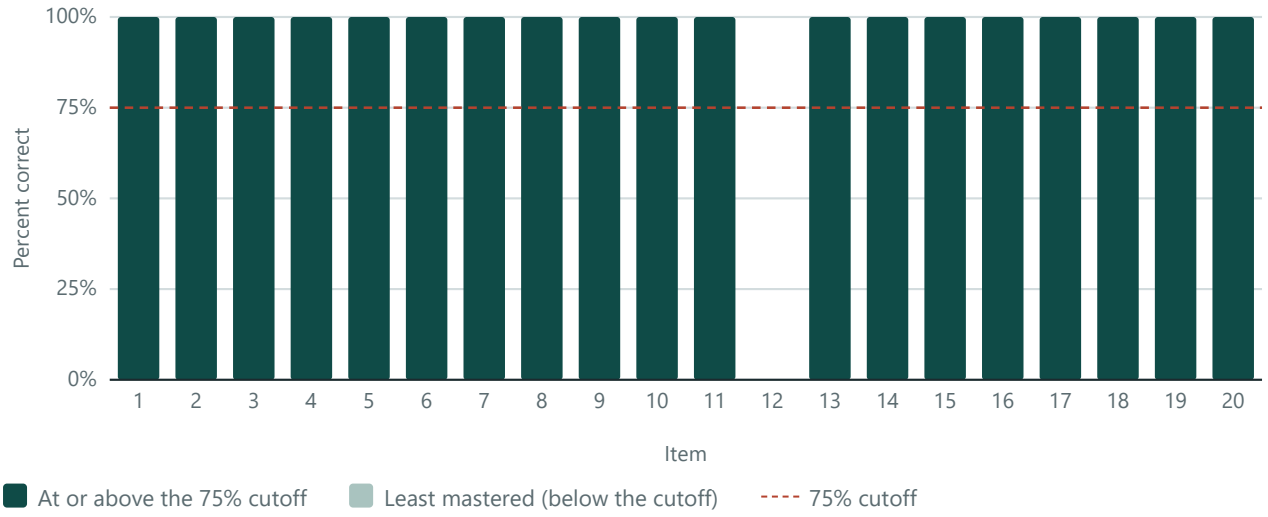



**Table 3.** Least Mastered Items (Below 75 Percent)

| Rank | Item | Correct (f) | Percentage | Mastery Level |
|---|---|---|---|---|
| 1 | 12 | 0 | 0.00 | Absolutely No Mastery |

**Table 4.** Most Mastered Items

| Item | Correct (f) | Percentage | Mastery Level |
|---|---|---|---|
| 1 | 1 | 100.00 | Mastered |
| 2 | 1 | 100.00 | Mastered |
| 3 | 1 | 100.00 | Mastered |
| 4 | 1 | 100.00 | Mastered |
| 5 | 1 | 100.00 | Mastered |

| Item | Correct (f) | Percentage | Mastery Level |
|---|---|---|---|
| 6 | 1 | 100.00 | Mastered |
| 7 | 1 | 100.00 | Mastered |
| 8 | 1 | 100.00 | Mastered |
| 9 | 1 | 100.00 | Mastered |
| 10 | 1 | 100.00 | Mastered |

### Interpretation

The reliability of the test could not be computed because every learner obtained the same score.

In terms of difficulty, the 20 items are 19 very easy and 1 very difficult. Based on both the difficulty and discrimination indices, 0 can be retained, 0 should be revised and 20 should be rejected.

Using the 75 percent cutoff, 1 of the 20 items are least mastered: items 12.

Because only 1 learner took the test, each 27 percent group holds 1 learner; read the discrimination indices with caution.

### Recommendations

1. Reteach the competencies measured by the least mastered items (items 12) before the next assessment.
2. Rewrite or replace the 20 items marked Revise or Reject before this test is used again, and keep the items marked Retain for the item bank.

## Notes on Computation

- Difficulty index (p) = learners who answered the item correctly ÷ all learners.
- Discrimination index (D) = (correct in the upper 27% − correct in the lower 27%) ÷ learners in one group (DepEd National Achievement Test criteria).
- Readings of p (0.81–1.00 Very Easy; 0.61–0.80 Easy; 0.41–0.60 Average; 0.21–0.40 Difficult; 0.00–0.20 Very Difficult), of D (0.40 and above Very Good; 0.30–0.39 Good; 0.20–0.29 Fair; below 0.20 Poor) and the decision (Retain when both are acceptable, Revise when one is, Reject when neither) follow standard test-construction practice.
- Reliability uses the Kuder-Richardson Formula 20 (KR-20). The upper and lower groups and KR-20 count correct answers, one per item, even when items carry different points.
- Mastery of an item uses the DepEd National Achievement Test criteria: 96–100 Mastered; 86–95 Closely Approximating Mastery; 66–85 Moving Towards Mastery; 35–65 Average Mastery; 15–34 Low Mastery; 5–14 Very Low Mastery; 0–4 Absolutely No Mastery.
- This report records results only; it does not compute or replace grades.

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